

Laplace Transform

CHAPTER HIGHLIGHTS

- ☞ Laplace Transforms
- ☞ The Region of Convergence
- ☞ Properties of ROC
- ☞ Laplace Transforms of Some Elementary Signals
- ☞ Properties of Laplace Transform
- ☞ Unilateral Laplace Transform
- ☞ Analysis of Continuous Time LTI Systems
- ☞ Characteristics of LTI Systems
- ☞ System Function for LTI Systems Described by Linear Constant-coefficient Differential Equations
- ☞ Systems Interconnections

LAPLACE TRANSFORMS

A continuous time LTI system with impulse response $h(t)$, and the output $y(t)$ of the system to the complex exponential input of the form e^{st} is $y(t) = h(t) * e^{st}$

$$= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) \cdot e^{-s\tau} \cdot d\tau$$

$$y(t) = T\{e^{st}\} = H(s)e^{st}$$

where $H(s) = \int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau$

The function $H(s)$ in the abovementioned equation is referred to as the Laplace transform of $h(t)$.

For a general continuous-time signal $x(t)$, the Laplace transform $X(s)$ is defined as $X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$ (bilateral Laplace transform)

$$X(s) = \int_0^{\infty} x(t) e^{-st} dt \quad (\text{unilateral Laplace transform})$$

both unilateral and bilateral are same, if $x(t) = 0$, for $t < 0$.

Symbolically, the transform will be represented by $X(s) = \mathcal{L}\{x(t)\}$

REGION OF CONVERGENCE

The range of values of the complex variables 's' for which the Laplace transform converges is called the region of convergence (ROC).

Consider the signal $x(t) = e^{-at}u(t)$, then Laplace transform

$$X(s) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+a)t} dt$$

$$= \frac{-1}{s+a} e^{-(s+a)t} \Big|_0^{\infty} = \frac{1}{s+a}, \quad \text{Re}(s) > -a$$

because $\lim_{t \rightarrow \infty} e^{-(s+a)t} = 0$, when $\text{Re}(s+a) > 0$ or $\text{Re}(s) > -a$ is the ROC.

The convenient way to display the ROC is shown in the figure, and variable s is a complex number; in the figure, we display the complex plane, generally referred to as the s -plane, the horizontal and vertical axes are sometimes referred to as $\text{Re}\{s\}$ or σ -axis and the $\text{Im}\{s\}$ or $j\omega$ -axis, respectively.

The shaded region in the figure represents the set of points in the S -plane corresponding to the region of convergence.

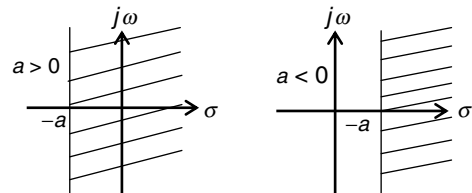


Figure 1 ROC for $X(s)$.

The ROC of $X(s)$ is $\text{Re}(s) > -a$, and this fact means that the integral defining $X(s)$ exists, only for the values of s in the shaded region; for other values of s , the integral does not converge. For this reason, the shaded region is called ROC (or) region of existence of $X(s)$.

Poles and Zeroes of $X(s)$

Usually, $X(s)$ will be a rational function in s , that is, $X(s)$

$$= \frac{a_0 (s - z_1)(s - z_2) \dots (s - z_m)}{b_0 (s - p_1)(s - p_2) \dots (s - p_n)}$$

where m and n are positive integers; $X(s)$ is proper rational function, if $n > m$; the roots of numerator polynomial z_k are called the zeroes of $X(s)$, as $X(s) = 0$ for those values. Similarly, the roots of denominator p_k are called the poles of $X(s)$ as for those values $X(s)$ is infinite.

Therefore, the poles of $X(s)$ lie outside of ROC, since $X(s)$ does not converge at the poles; however, zeroes may lie inside or outside of ROC.

Properties of ROC

1. The ROC of $X(s)$ consists of strips parallel to the $j\omega$ -axis in the s -plane and ROC does not contain any poles.
2. If $x(t)$ is finite-duration signal and absolutely integrable, that is, $x(t) \neq 0$, $t_1 \leq t \leq t_2 = 0$, otherwise; Then, ROC is the entire s -plane except possibly $s = 0$ or $s = \infty$.
3. If $x(t)$ is right-hand side signals $x(t) = 0$ for $t < t_1 < \infty$, then ROC is of the form $\text{Re}(s) > \sigma_{\max}$, where $\sigma_{\max}$ equals the maximum real part of any poles of $X(s)$. Thus, ROC is to the right of all the poles of $X(s)$.
4. If $x(t)$ is a left-hand side signal, that is, $x(t) = 0$, for $t > t_2 > -\infty$, then ROC is of the form $\text{Re}(s) < \sigma_{\min}$, where $\sigma_{\min}$ is the minimum real part of any of the poles of $X(s)$. Thus, ROC is to the left of all the poles of $X(s)$.
5. If $x(t)$ is a two-sided signal, that is, $x(t)$ is an infinite-duration signal, then ROC is of the form $\sigma_1 < \text{Re}(s) < \sigma_2$ where σ_1 and σ_2 are real parts of the two poles of $X(s)$. Thus, ROC is a vertical strip in the s -plane between σ_1 and σ_2 .

Solved Examples

Example 1

Find the Laplace transform of $x(t) = e^{-b|t|}$

Solution

$x(t)$ be expressed as $x(t)$.

$$= e^{-bt} u(t) + e^{+bt} u(-t)$$

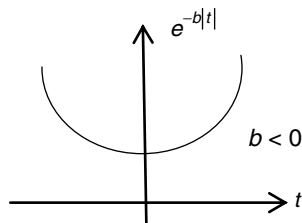
Therefore, the Laplace transform of

$$e^{-bt} u(t) \leftrightarrow \frac{1}{s+b} \quad \text{Re}\{s\} > -b$$

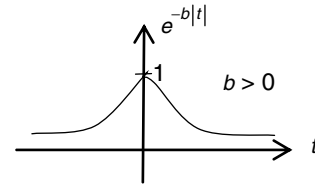
Further, for $e^{+bt} u(-t) \leftrightarrow \frac{-1}{s-b}$, $\text{Re}\{s\} < +b$

If $b \leq 0$, then there is no common region of ROC.

Thus, for those values of b , $x(t)$ has no Laplace transform.



For $b < 0$, the signal is also not finite integrable, and therefore, no Laplace transform exists.

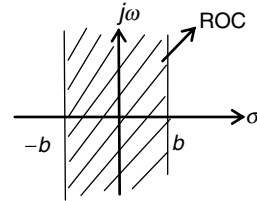


For $b > 0$, the signal is integrable, and there is common region of convergence, that is, $-b < \text{Re}\{s\} < +b$

$$e^{-b|t|} \xrightarrow{L} \frac{1}{s+b} - \frac{1}{s-b} = \frac{-2b}{s^2 - b^2}$$

$$-b < \text{Re}\{s\} < +b$$

(The corresponding pole-zero plot is)



LAPLACE TRANSFORMS OF SOME ELEMENTARY SIGNALS

1. Unit impulse function $\delta(t)$

$$L\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t) dt = 1, \text{ all } s$$

$$L\{\delta(t-T)\} = e^{-sT}, \text{ all } s.$$

2. Unit step function $u(t)$

$$\begin{aligned} L[u(t)] &= \int_{-\infty}^{\infty} u(t) e^{-st} dt = \int_0^{\infty} e^{-st} dt \\ &= \frac{1}{s} e^{-st} \Big|_0^{\infty} = \frac{1}{s} \quad \text{Re}(s) > 0 \end{aligned}$$

3. Some other Laplace transform pairs

$$L\{-u(-t)\} = \frac{1}{s}, \quad \text{Re}(s) < 0$$

$$L\{tu(t)\} = \frac{1}{s^2}, \quad \text{Re}(s) > 0$$

$$L\{t^k u(t)\} = \frac{k!}{s^{k+1}}, \quad \text{Re}(s) > 0$$

$x(t)$	$X(s)$	ROC
$e^{-at} u(t)$	$\frac{1}{s+a}$	$\text{Re}(s) > -a$
$-e^{-at} u(-t)$	$\frac{1}{s+a}$	$\text{Re}(s) < -a$
$te^{-at} u(t)$	$\frac{1}{(s+a)^2}$	$\text{Re}(s) > -a$
$-te^{-at} u(-t)$	$\frac{1}{(s+a)^2}$	$\text{Re}(s) < -a$
$\cos \omega_0 t u(t)$	$\frac{s}{s^2 + \omega_0^2}$	$\text{Re}(s) > 0$

(Continued)

$x(t)$	$X(s)$	ROC
$\sin \omega_0 t u(t)$	$\frac{\omega_0}{s^2 + \omega_0^2}$	$\text{Re}(s) > 0$
$e^{-at} \cos \omega_0 t u(t)$	$\frac{s+a}{(s+a)^2 + \omega_0^2}$	$\text{Re}(s) > -a$
$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(s+a)^2 + \omega_0^2}$	$\text{Re}(s) > -a$
$t^k e^{-at} u(t)$	$\frac{k!}{(s+a)^{k+1}}$	$\text{Re}(s) > -a$
$\frac{d^n}{dt^n} \delta(t)$	s^n	All s
$u(t) * u(t) * \dots * n$ times	$1/s^n$	$\text{Re}(s) > 0$

PROPERTIES OF LAPLACE TRANSFORM

1. Linearity

Laplace transform is a linear operator. It holds the principle of superposition.

If $L\{x_1(t)\} = X_1(s)$, $\text{ROC} = R_1$

$L\{x_2(t)\} = X_2(s)$, $\text{ROC} = R_2$

$L\{a_1 x_1(t) + a_2 x_2(t)\} = a_1 X_1(s) + a_2 X_2(s)$, $\text{ROC} = R_1 \cap R_2$

2. Time shifting

The time shifting property states that delaying a signal by t_0 seconds amounts to multiplying its transform with e^{-st_0} .

$L\{x(t)\} = X(s)$, $\text{ROC} = R$

$L\{x(t - t_0)\} = e^{-st_0} X(s)$, $\text{ROC} = R$

3. Shifting in s -domain

Shifting in s -domain or frequency shifting property states that frequency shift by s_0 is same as multiplying its inverse transform with $e^{s_0 t}$.

$e^{s_0 t} x(t) \leftrightarrow X(s - s_0)$, $\text{ROC} = R + \text{Re}(s_0)$

If the Laplace transform of $x(t)$ has a pole or zero at $s = a$, then the Laplace transform of $e^{j\omega_0 t} x(t)$ has a pole or zero at $s = a + j\omega_0$.

4. Time scaling

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{s}{a}\right), \text{ROC} = R/a$$

$x(at)$ is the signal $x(t)$ and time compressed by the factor a , and $X\left(\frac{s}{a}\right)$ is $X(s)$ expanded along the s -scale by the same factor a .

The scaling property states that time compression of a signal by a factor 'a' causes expansion of its Laplace transform in the s -scale by the same factor.

Similarly, time expansion $x(t)$ causes compression of $X(s)$ in the s -scale by the same factor.

5. Time reversal:

$$x(-t) \leftrightarrow X(-s), \text{ROC} = -R$$

6. Differentiation in time domain

$$\frac{d}{dt} [x(t)] \leftrightarrow sX(s) - x(0^-), \text{ROC} \supset R$$

Repeated application of this property yields

$$\frac{d^2 x(t)}{dt^2} \leftrightarrow s^2 X(s) - sx(0^-) - \frac{d}{dt} x(0^-)$$

$$\frac{d^n x(t)}{dt^n} \leftrightarrow s^n X(s) - \sum_{k=1}^n s^{n-k} \frac{d^{k-1} x(0^-)}{dt^{k-1}}$$

7. Differentiation in the s -domain.

The dual of the time differentiation property is the frequency differentiation property.

$$-t x(t) \leftrightarrow \frac{d}{ds} X(s), R^1 = R$$

8. Integration in the time domain

$$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{s} X(s) + \frac{1}{s} \int_{-\infty}^{0^-} x(t) dt,$$

$$\text{ROC} = R \cap \{\text{Re}(s) > 0\}$$

9. Time Convolution

$$x_1(t) * x_2(t) \leftrightarrow X_1(s) X_2(s), \text{ROC} \supset R_1 \cap R_2$$

We can apply the time convolution property to the LTI system input-output relationship $y(t) = x(t) * h(t)$ to obtain $Y(s) = X(s) H(s)$.

$$H(s) = \frac{Y(s)}{X(s)} = \text{transfer function } H(s).$$

10. Time differentiation

$$\frac{d}{dt} x(t) \leftrightarrow sX(s) - x(0^-)$$

$$\frac{d^2}{dt^2} x(t) \leftrightarrow s^2 X(s) - sx(0^-) - x'(0^-)$$

$$\frac{d^n}{dt^n} x(t) \leftrightarrow s^n X(s) - s^{n-1} x(0^-) - s^{n-2} x'(0^-) - \dots - sx^{(n-2)}(0^-) - x^{(n-1)}(0^-)$$

11. Frequency integration

The dual of the time-integration property is the frequency-integration property.

$$\frac{x(t)}{t} \leftrightarrow \int_s^\infty X(p) dp$$

12. Frequency convolution

$$x_1(t) x_2(t) \leftrightarrow \frac{1}{2\pi j} [X_1(s) * X_2(s)]$$

13. Initial value theorem

$$x(0^+) = \lim_{s \rightarrow \infty} sX(s)$$

14. Final value theorem

$$\lim_{s \rightarrow 0} sX(s) = x(\infty)$$

It is desirable to know the values of $x(t)$ as $t \rightarrow 0$ and $t \rightarrow \infty$ [initial and final values of $x(t)$] from the knowledge of its Laplace transform $X(s)$, then initial and final value theorems provide such information.

15. Conjugation property:

$$x(t) \leftrightarrow X(s) \text{ with } \text{ROC} = R, \text{ then}$$

Conjugate of $x(t)$, $x^*(t) \leftrightarrow X^*(s^*)$ with $\text{ROC} = R$

When $x(t)$ is real, $x(t) = x^*(t)$, and therefore,
 $X(s) = X^*(s^*)$.

Consequently, if $x(t)$ is real, and if $X(s)$ has a pole/zero at $s = s_0$, then $X(s)$ also has pole or zero at the complex point $s = s_0^*$.

INVERSE LAPLACE TRANSFORM

$$x(t) \leftrightarrow X(s)$$

$$x(t) = L^{-1}\{X(s)\}$$

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st} ds$$

In this integral, the real ' σ ' is to be selected such that if the ROC of $X(s)$ is $\sigma_1 < \text{Re}(s) < \sigma_2$, then $\sigma_1 < \sigma < \sigma_2$.

UNILATERAL LAPLACE TRANSFORM

The Laplace transforms for the signals $e^{-at}u(t)$ and $-e^{-at}u(-t)$ are identical except for their regions of convergence. Therefore, for a given $X(s)$, there may be more than one inverse transform, depending on the ROC. If we restrict all our signals to the causal type, such an ambiguity does not arise. There is only one inverse transform of $X(s) = \frac{1}{(s+a)}$ namely $e^{-at}u(t)$. To find $x(t)$ from $X(s)$, we need not even specify ROC. If all signals are restricted to the causal type, then for a given $X(s)$, there is only one inverse Laplace transform $x(t)$.

The unilateral Laplace transform is a special case of the bilateral Laplace transform in which all signals are restricted to being causal.

The unilateral Laplace transform $X(s)$ of $x(t)$ is $X(s) = \int_0^{\infty} x(t)e^{-st} dt$

Example 2

What is the Laplace transform of the signal $x(t) = e^{-3t}u(t) + e^{-t}(\cos 2t)u(t)$?

Solution

$$x(t) = \{e^{-3t} + \frac{1}{2}(e^{2jt} + e^{-2jt}) \cdot e^{-t}\}u(t)$$

$$= e^{-3t}u(t) + \frac{1}{2}e^{(-1+2j)t}u(t) + \frac{1}{2}e^{(-1-2j)t}u(t)$$

By taking Laplace transform,

$$X(s) = \frac{1}{s+3} + \frac{1}{2} \cdot \frac{1}{s+(1+2j)} + \frac{1}{2} \cdot \frac{1}{s+(1-2j)}$$
 and the ROC

is $\text{Re}\{s\} > -3$ for the first term, and for second and third terms $\text{Re}\{s\} > -1$

For all three Laplace transforms to converge simultaneously, we must have $\text{Re}\{s\} > -1$

$$\text{Consequently, } X(s) = \frac{1}{s+3} + \frac{1}{2} \cdot \frac{2s+2}{s^2+2s+5}$$

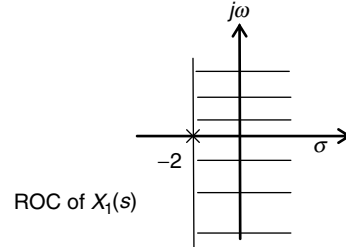
$$= \frac{1}{s+3} + \frac{s+1}{s^2+2s+5} \quad \text{Re}\{s\} > -1$$

Example 3

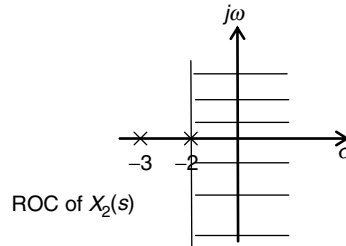
Consider $X_1(s) = \frac{1}{s+2}$ $\text{Re}\{s\} > -2$ and $X_2(s) = \frac{1}{(s+2)(s+3)}$, $\text{Re}\{s\} > -2$. Then, what is the ROC of $X(s) = X_1(s) - X_2(s)$?

Solution

$$X_1(s) = \frac{1}{s+2} \quad \text{Re}\{s\} > -2$$



$$X_2(s) = \frac{1}{(s+2)(s+3)} \quad \text{Re}\{s\} > -2$$



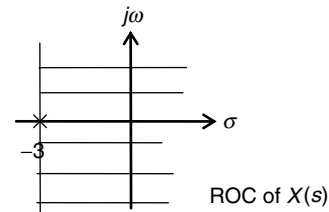
$$X(s) = X_1(s) - X_2(s)$$

$$= \frac{1}{s+2} - \frac{1}{(s+2)(s+3)} = \frac{s+3-1}{(s+2)(s+3)}$$

$$= \frac{(s+2)}{(s+2)(s+3)} = \frac{1}{s+3}, \quad \text{Re}\{s\} > -3$$

Thus, in the linear combination of $X_1(s)$ and $X_2(s)$, the pole at $s = -2$ is cancelled by a zero at $s = -2$. The intersection of the ROC for $X_1(s)$ and $X_2(s)$ is $\text{Re}\{s\} > -2$.

However, since the ROC is always bounded by a pole or infinity, for this example, the ROC of $X(s)$ can be extended to the left to be bounded by the pole at $s = -3$, as a result of the pole-zero cancellation at $s = -2$.



Example 4

What is the inverse Laplace transform of $X(s) = \frac{2s^2+5}{s^2+3s+2}$?

Solution

$$X(s) = \frac{2s^2+5}{s^2+3s+2} = 2 + \frac{A}{(s+1)} + \frac{B}{(s+2)},$$

where $A = (s+1)X(s) \Big|_{s=-1} = \frac{2+5}{-1+2} = 7$

$$B = (s+2)X(s) \Big|_{s=-2} = \frac{8+5}{-2+1} = -13$$

Therefore, $X(s) = 2 + \frac{7}{s+1} - \frac{13}{s+2}$

By taking inverse Laplace transforms

$$x(t) = 2\delta(t) + 7e^{-t}u(t) - 13e^{-2t}u(t)$$

Example 5

Find the inverse Laplace transform of

$$X(s) = \frac{s+2+4e^{-3s}}{(s+3)(s+1)}$$

Solution

Observe the exponential term e^{-3s} in numerator of $X(s)$, indicating time delay.

$$\begin{aligned} X(s) &= \frac{s+2}{(s+3)(s+1)} + \frac{4e^{-3s}}{(s+3)(s+1)} \\ &= X_1(s) + X_2(s) e^{-3s} \end{aligned}$$

Therefore, $X_1(s) = \frac{s+2}{(s+1)(s+3)} = \frac{1}{2} \frac{1}{s+1} + \frac{1}{2} \frac{1}{s+3}$

By applying inverse Laplace transform

$$x_1(t) = \frac{1}{2} e^{-t} u(t) + \frac{1}{2} e^{-3t} u(t)$$

Similarly, $X_2(s) = \frac{4}{(s+1)(s+3)} = \frac{2}{s+1} - \frac{2}{s+3}$

By applying inverse Laplace transform

$$x_2(t) = 2e^{-t} u(t) - 2e^{-3t} u(t)$$

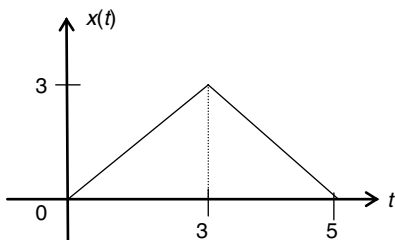
$$X(s) = X_1(s) + X_2(s) e^{-3s}$$

Therefore, $x(t) = x_1(t) + x_2(t-3)$

$$= \frac{1}{2} [e^{-t} + e^{-3t}] u(t) + 2[e^{-(t-3)} - e^{-3(t-3)}] u(t-3)$$

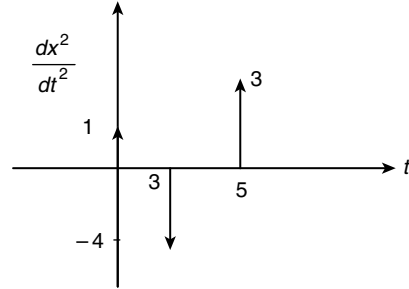
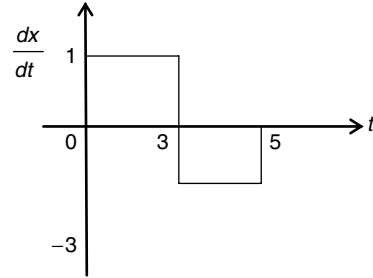
Example 6

Find the Laplace transform of the signal $x(t)$ depicted in the figure.



Solution

By considering the derivative of $x(t)$



$$\Rightarrow \frac{d^2x}{dt^2} = \delta(t) - 4\delta(t-3) + 3\delta(t-5)$$

By applying Laplace transform, $L\left\{\frac{d^2x}{dt^2}\right\}$

$$= s^2X(s) = 1 - 4e^{-3s} + 3e^{-5s}$$

$$X(s) = \frac{1}{s^2}(1 - 4e^{-3s} + 3e^{-5s})$$

ANALYSIS OF CONTINUOUS TIME LTI SYSTEMS

System Function

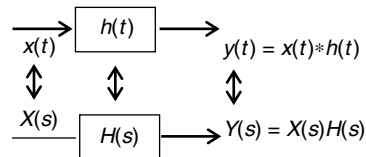
The output of C.T. LTI system equals to the convolution of the input $x(t)$ with the impulse response $h(t)$, that is,

$$y(t) = x(t) * h(t)$$

By applying transform

$$Y(s) = X(s) H(s)$$

and $H(s) = \frac{Y(s)}{X(s)}$



The Laplace transform $H(s)$ of $h(t)$ is referred to as the system function (or the transfer function) of the system.

Characteristics of LTI Systems

Many properties of continuous time LTI systems can be closely associated with the characteristics of $H(s)$ in s -plane and ROC.

Causality

For causal continuous time LTI system, we have $h(t) = 0$, $t < 0$. Since $h(t)$ is right-hand side signal, the corresponding $H(s)$ will have ROC $\text{Re}(s) > \sigma_{\max}$.

In other words, for a system with a rational system function, the causality of the system is equivalent to the ROC being the right half plane to the right of the right-most pole.

Similarly, for anti-causal $h(t) = 0$, $t > 0$, and $H(s)$ ROC will be $\text{Re}(s) < \sigma_{\min}$ (left-half plane).

Stability

A continuous time LTI system is BIBO stable if $\int_{-\infty}^{\infty} |h(t)| dt < \infty$.

The corresponding $H(s)$, ROC contains the $j\omega$ -axis or an LTI system is stable if and only if the ROC of its system function $H(s)$ includes the $j\omega$ -axis (i.e., $\text{Re}\{s\} = 0$).

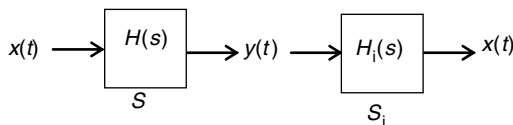
Causal and Stable System

If the system is both causal and stable, then the poles of $H(s)$ must lie in the left half of the s -plane.

Since ROC is of form $\text{Re}(s) > \sigma_{\max}$, and $j\omega$ -axis is included in ROC, we must have $\sigma_{\max} < 0$. In other terms, a causal system with rational system function $H(s)$ is stable if and only if all of the poles of $H(s)$ lie in the left half of the s -plane, that is, all of the poles have negative real parts.

Invertibility

If $H(s)$ is the transfer function of a system S , then S_i its inverse system has a transfer function. $H_i(s)$ given by $H_i(s) = \frac{1}{H(s)}$



The cascade of systems S and S_i is an identity system with impulse response $\delta(t)$.

Implying $H(s)H_i(s) = 1$

Consider an ideal differentiator system $y(t) = \frac{d}{dt}x(t)$ and its Laplace transform $Y(s) = sX(s)$ (for causal system $x(0^-) = 0$)

Therefore, $H(s) = \frac{Y(s)}{X(s)} = s$ is the transfer function.

For an ideal integrator system with zero initial state

$$y(t) = \int_0^t x(\tau) d\tau \rightarrow Y(s) = \frac{1}{s} X(s)$$

$$\text{Transfer function } H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s}$$

From the abovementioned discussion, an ideal integrator and its inverse and an ideal differentiator have transfer functions $1/s$ and s , respectively, leading to $H(s)$. $H_i(s) = 1$.

Example 7

Check the causality of the system function $H(s) = \frac{e^s}{s+1}$ $\text{Re}\{s\} > -1$

Solution

The ROC is to the right of the right-most pole. Therefore, impulse response must be one-sided.

$$e^{-t} u(t) \leftrightarrow \frac{1}{s+1} \text{Re}\{s\} > -1$$

By using time shift property

$$x(t - t_0) \leftrightarrow e^{-t_0 s} X(s)$$

$$e^{-(t+1)} u(t+1) \leftrightarrow \frac{e^s}{s+1} \text{Re}\{s\} > -1$$

Therefore, impulse response $h(t) = e^{-(t+1)} u(t+1)$, which is non-zero for $-1 < t < 0$, and hence, the system is not causal.

NOTE

The causality implies that the ROC is to the right of the right-most pole, but the converse is not in general true, unless the system function is rational.

Example 8

Consider an LTI system with systemfunction $H(s) = \frac{s-1}{(s+2)(s-3)}$. Check the stability and causality for different ROCs.

Solution

$$H(s) = \frac{s-1}{(s+2)(s-3)} = \frac{3}{5} \cdot \frac{1}{s+2} + \frac{2}{5} \cdot \frac{1}{s-3}$$

Possible ROC are $\text{Re}\{s\} > 3$, $\text{Re}\{s\} < -2$, and $-2 < \text{Re}\{s\} < 3$.

When ROC is $\text{Re}\{s\} > 3$

$$\begin{aligned} h(t) &= \frac{3}{5} e^{-2t} u(t) + \frac{2}{5} e^{+3t} u(t) \\ &= \frac{1}{5} \{3e^{-2t} + 2e^{3t}\} u(t) \end{aligned}$$

The system is causal and unstable

When ROC is $-2 < \text{Re}\{s\} < 3$

$$\text{Re}\{s\} < 3 \rightarrow -e^{3t} u(-t) \leftrightarrow \frac{+1}{s-3}$$

$$\text{Re}\{s\} > -2 \rightarrow e^{-2t} u(t) \leftrightarrow \frac{1}{s+2}$$

$$\text{Therefore, } h(t) = \frac{3}{5} \cdot e^{-2t} u(t) - \frac{2}{5} e^{3t} u(-t)$$

The system is non-causal and stable.

When ROC is $\text{Re}\{s\} < -2$

$$h(t) = -\frac{3}{5} \cdot e^{-2t} u(-t) - \frac{2}{5} e^{3t} u(-t)$$

The system is anti-causal and unstable.

System Function for LTI Systems Described by Linear Constant-coefficient Differential Equations

An LTI system with input $x(t)$, and the output of $y(t)$ in differential equation form

$$\sum_{k=0}^N a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^M b_k \frac{d^k}{dt^k} x(t)$$

Applying the Laplace transform and using differentiation property

$$\sum_{k=0}^N a_k s^k Y(s) = \sum_{k=0}^M b_k s^k X(s) \big|_{\text{Zero initial condition}}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{\sum_{k=0}^M b_k s^k}{\sum_{k=0}^N a_k s^k}$$

Example 9

Consider an LTI system for which the input $x(t)$ and output $y(t)$ satisfy the linear constant co-efficient differential equation.

$\frac{dy(t)}{dt} + 2y(t) = x(t)$. With zero initial conditions, find the impulse response of the system (1) when it is causal system? and (2) when it is anti-causal system?

Solution

$$\frac{dy(t)}{dx} + 2y(t) = x(t)$$

By applying Laplace transform,

$$sY(s) + 2Y(s) = X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+2}$$

The abovementioned expression is system function, and however, we do not know the ROC of the system. When the system is causal, the ROC can be inferred to be to the right of the right-most pole, which this case corresponds to $\text{Re}\{s\} > -2$.

Then, $h(t) = e^{-2t} u(t)$

When the system is anti-causal, the ROC associated with $H(s)$ would be $\text{Re}\{s\} < -2$. Then, $h(t) = -e^{-2t} u(-t)$.

Example 10

If input to an LTI system is $x(t) = e^{-2t} u(t)$, then the output is $y(t) = [e^{-t} - e^{-3t}] u(t)$.

What is the transfer function of the system and the differential equation of the system with zero initial conditions?

Solution

By taking Laplace transform for $x(t)$ and $y(t)$, we get $X(s) =$

$$\frac{1}{s+2} \quad \text{Re}\{s\} > -2$$

$$\text{and } Y(s) = \frac{1}{s+1} - \frac{1}{s+3} \quad \text{Re}\{s\} > -1$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{2/(s+1)(s+3)}{1/s+2} = \frac{2(s+2)}{(s+1)(s+3)}$$

According to the convolution property, the ROC of $Y(s)$ must include at least the intersections of the ROCs of $X(s)$ and $H(s)$.

Therefore, $H(s)$ ROC should be $\text{Re}\{s\} > -1$ and $H(s)$ is causal.

Since both poles of $H(s)$ have negative real parts, it follows that the system is stable.

$$\text{We can write } \frac{Y(s)}{X(s)} = \frac{2(s+2)}{(s+1)(s+3)} = \frac{2s+4}{s^2+4s+3}$$

$$s^2 Y(s) + 4s Y(s) + 3 Y(s) = 2s X(s) + 4 X(s)$$

$\frac{d^2}{dt^2} y(t) + 4 \frac{dy(t)}{dt} + 3y(t) = 2 \frac{dx(t)}{dt} + 4x(t)$ is the differential equation of the system specified.

Example 11

From the following information about an LTI system, find the system function of that LTI system.

- (1) The system is causal.
- (2) The system function is rational and has only one pole at $s = -1$ and $s = 3$.
- (3) if $x(t) = 1$, then output $y(t) = 0$.
- (4) The value of the impulse response at $t = 0^+$ is 2.

Solution

Consider the system function

$$H(s) = \frac{P(s)}{Q(s)}, \text{ and } Q(s) = (s+1)(s-3).$$

$$H(s) = \frac{P(s)}{(s+1)(s-3)}$$

$y(t)$ is the output of the system for input $x(t) = e^{st}$, then $y(t) = H(s) e^{st}$. Here, for $x(t) = 1 = e^{0t}$.

$y(t) = H(0)e^{0t} = 0$, $H(0) = 0$, which means that $H(s)$ must have a zero at $s = 0$. Therefore, $P(s) = sP^1(s)$.

The value of impulse response at

$$t = 0 + h(t) \Big|_{t=0^+} = \lim_{s \rightarrow \infty} s H(s) = \lim_{s \rightarrow \infty} \frac{s^2 P^1(s)}{s^2 - 2s - 3} = 2$$

Therefore, $P^1(s) = 2$, that is, it is a constant and $H(s) = \frac{2s}{(s+1)(s-3)}$

Further, the system is unstable as it has a pole at $s = 3$ with positive real part.

Example 12

A system is described by the differential equation

$$\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 3y(t) = x(t).$$

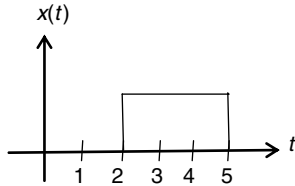
Let $x(t)$ be a rectangular pulse given by

$$x(t) = \begin{cases} 1 & 2 < t < 5 \\ 0 & \text{otherwise} \end{cases}$$

Assuming zero initial conditions, find the Laplace transform of $y(t)$.

Solution

$$x(t) = \begin{cases} 1 & 2 < t < 5 \\ 0 & \text{otherwise} \end{cases}$$



$$x(t) = u(t-2) - u(t-5)$$

By taking Laplace transform,

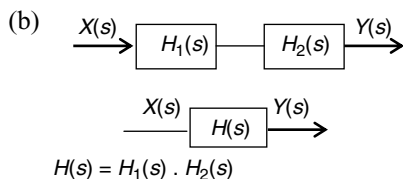
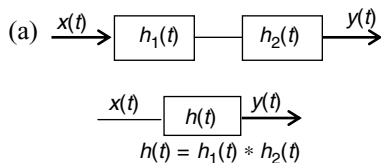
$$X(s) = \frac{e^{-2s}}{s} - \frac{e^{-5s}}{s}$$

For the given differential equation, by considering Laplace transform $s^2 Y(s) + 4s Y(s) + 3 = X(s)$.

$$Y(s) = \frac{X(s)}{s^2 + 4s + 3} = \frac{e^{-2s} - e^{-5s}}{s(s+1)(s+3)}$$

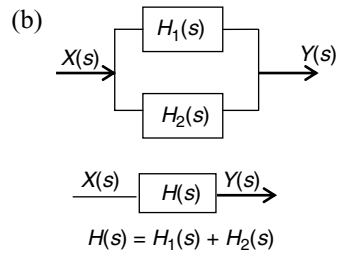
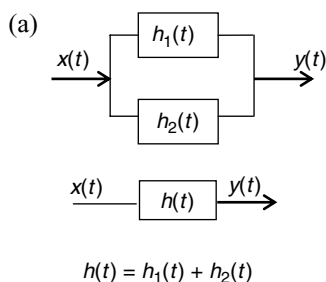
Systems' Interconnections

Two systems in cascade,



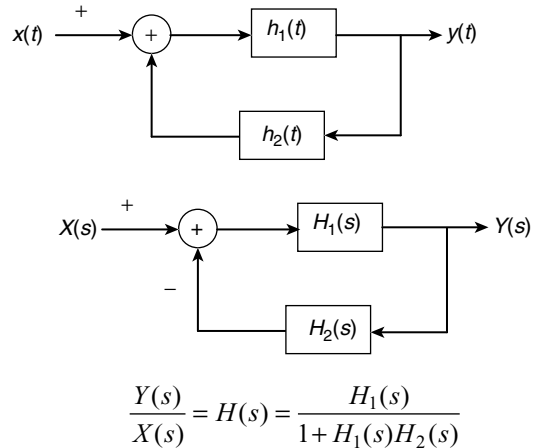
(a) Time domain, (b) s-domain

Two systems are in parallel



(a) Time domain, (b) s-domain

Feedback interconnection of two LTI systems.



Example 13

$X(s) = e^{-s} \left[\frac{-2}{s(s+2)} \right]$. What are the initial and final values of $x(t)$?

- (A) -1 and 0 (B) 0 and 1
(C) 0 and -1 (D) 1 and 1

Solution

Initial value theorem

$$x(0^+) = \lim_{s \rightarrow \infty} s X(s) = \lim_{s \rightarrow \infty} s e^{-s} \left[\frac{-2}{s(s+2)} \right] = 0$$

Final value theorem $x(\infty) = \lim_{s \rightarrow 0} s X(s)$

$$= \lim_{s \rightarrow 0} s e^{-s} \frac{-2}{s(s+2)} = -1$$

Example 14

Using Laplace transform, find out output signal, when $h(t) = e^{-2t}u(t)$, $t > 0$

$$x(t) = e^{-6t}u(t) \quad t > 0$$

$$y(t) = ?$$

- (A) $\left(\frac{1}{4}e^{-2t} - \frac{1}{4}e^{-6t} \right) u(t) = y(t)$, $\text{Re}(s) > -2$
(B) $\left(\frac{1}{4}e^{-2t} + \frac{1}{4}e^{-6t} \right) u(t) = y(t)$, $\text{Re}(s) > 2$

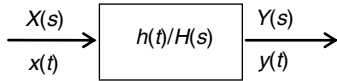
$$(C) \left(-\frac{1}{4}e^{-2t} - \frac{1}{4}e^{6t} \right) u(t) = y(t) \operatorname{Re}(s) > -2$$

$$(D) \left(\frac{1}{4}e^{-2t} - \frac{1}{4}e^{-6t} \right) u(t) = y(t) \operatorname{Re}(s) > -2$$

Solution

$$h(t) = e^{-2t} u(t)$$

$$x(t) = e^{-6t} u(t)$$



Time response output $y(t) = h(t) * x(t)$

Frequency response

$$Y(s) = X(s) H(s)$$

$$Y(s) = \frac{1}{s+6} \cdot \frac{1}{s+2} \operatorname{Re}(s) > -2 \text{ (or) } \sigma > -2$$

$$Y(s) = \frac{A}{s+6} + \frac{B}{s+2}$$

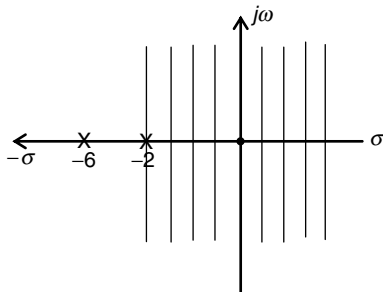
By using factorial method,

$$A = \lim_{s \rightarrow -6} (s+6)Y(s) = \lim_{s \rightarrow -6} (s+6) \frac{1}{(s+2)(s+6)} = \frac{-1}{4}$$

$$B = \lim_{s \rightarrow -2} (s+2) \frac{1}{(s+2)(s+6)} = \frac{1}{4}$$

$$Y(s) = \frac{-1}{4(s+6)} + \frac{1}{4(s+2)}, \sigma > -2$$

$$y(t) = \left(\frac{1}{4}e^{-2t} - \frac{1}{4}e^{-6t} \right) u(t),$$



Right-most signal
stable and causal

Example 15

If Laplace transform of a signal

$$Y(s) = \frac{1}{s(s-1)}, \text{ then its final value is}$$

$$(A) 1 \quad (B) -1$$

$$(C) 0 \quad (D) \infty$$

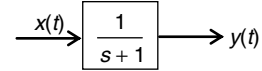
Solution

Final value theorem

$$y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} \frac{s}{s(s-1)} = -1$$

Example 16

For the following system, $x(t) = (\sin t)u(t)$. In steady state, the response $y(t)$ will be



$$(A) \frac{1}{\sqrt{2}} \sin t$$

$$(B) \frac{1}{\sqrt{2}} \sin \left(t - \frac{\pi}{4} \right)$$

$$(C) \frac{1}{\sqrt{2}} \cos \left(t - \frac{\pi}{4} \right)$$

$$(D) \frac{1}{\sqrt{2}} \cos t$$

Solution

$$H(s) = \frac{1}{s+1}$$

$$x(t) = (\sin t)u(t) \leftrightarrow X(s) = \frac{1}{s^2+1}$$

$$Y(s) = H(s)X(s) = \frac{1}{(s+1)(s^2+1)}$$

$$= \frac{1}{2} \left[\frac{1}{s+1} - \frac{s-1}{s^2+1} \right]$$

$$= \frac{1}{2} \left[\frac{1}{s+1} - \frac{s}{s^2+1} + \frac{1}{s^2+1} \right]$$

$$y(t) = \frac{1}{2} [e^{-t} - \cos t + \sin t] u(t)$$

$$\text{at steady state, } y(t) = \frac{1}{2} [\sin t - \cos t] u(t)$$

$$= \frac{1}{\sqrt{2}} \sin \left(t - \frac{\pi}{4} \right)$$

Example 17

Unit step response of a system starting from rest is given by $c(t) = 1 - e^{-2t}$ for $t \geq 0$. The transfer function of the system is

$$(A) \frac{s}{s+2}$$

$$(B) \frac{1}{s+2}$$

$$(C) \frac{s}{s-2}$$

$$(D) \frac{2}{s+2}$$

Solution

Unit step response $c(t) = 1 - e^{-2t}$, for $t > 0$

Impulse response $h(t) = \frac{d}{dt} c(t) = 2e^{-2t}$ for $t > 0$. Therefore,

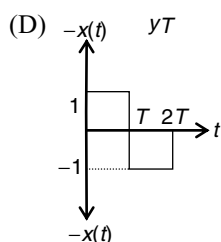
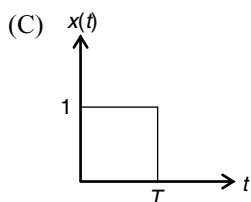
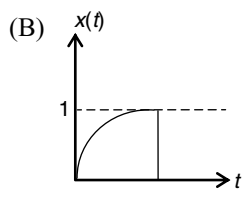
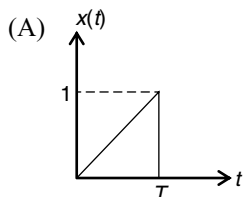
$$\text{transfer function } H(s) = \frac{2}{s+2}$$

EXERCISES

Practice Problems I

Direction for questions 1 to 22: Select the correct alternative from the given choices.

1. Inverse Laplace transform of $X(s)$ is $x(t)$. When $X(s) = \frac{1 - e^{-sT}}{s}$, shape of $x(t)$ is?



2. Match the following:

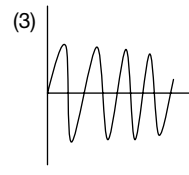
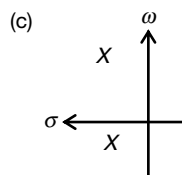
List-I	List-II
(a) $\frac{10}{s(s+10)}$	(1) $10 \delta(t)$
(b) $\frac{10}{s^2 + 100}$	(2) $(e^{-10t} \cos 10t)u(t)$
(c) $\frac{(s+10)}{(s+10)^2 + 100}$	(3) $(\sin 10t)u(t)$
(d) 10	(4) $(1 - e^{-10t})u(t)$

Codes:

- | | |
|-------------|-------------|
| a b c d | a b c d |
| (A) 3 4 1 2 | (B) 4 3 2 1 |
| (C) 3 4 1 2 | (D) 4 3 1 2 |

3. Match the following:

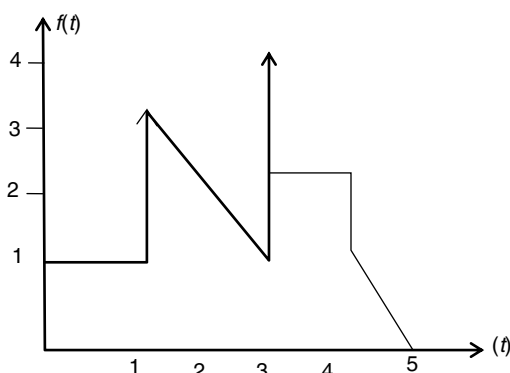
List-I	List-II
Pole-zero plot	Response
(a)	(1)
(b)	(2)



- | | |
|-----------|-----------|
| a b c | a b c |
| (A) 1 2 3 | (B) 3 2 1 |
| (C) 2 3 1 | (D) 3 1 2 |

Direction for questions 4 and 5:

4. Resolve $f(t)$ in terms of step, impulse, and ramp functions and also find out Laplace transform



- (A) $\frac{1}{s}[1 + 2e^{-s} + e^{-3s} - e^{-4s}] + \frac{1}{s^2}[e^{-s} - e^{-3s} + 4e^{-4s} - e^{-5s}] + 2e^{-3s}$
- (B) $\frac{1}{s}[1 + 2e^{-s} + e^{-3s} - e^{-4s}] + \frac{1}{s^2}[e^{-s} - e^{-3s} + e^{-4s} - e^{-5s}] + 2e^{-3s}$
- (C) $\frac{1}{s}[1 + 2e^{-s} + 3e^{-3s} - e^{-4s}] - \frac{1}{s^2}[e^{-s} - e^{-3s} + e^{-4s} - e^{-5s}] + 2e^{-3s}$
- (D) $\frac{1}{s}[1 + 2e^{-s} + e^{-3s} - e^{-4s}] - \frac{1}{s^2}[e^{-s} - e^{-3s} + e^{-4s} - e^{-5s}] - 2e^{-3s}$

5. From above mentioned figure, find out the initial value
(A) 0 (B) 4 (C) 2 (D) 1

Direction for questions 6 and 7:

The impulse response $h(t)$ of an LTI continuous time system is given by $h(t) = \exp(-2t)u(t)$, where $u(t)$ denotes the unit step function

(Continued)

6. The frequency response $H(\omega)$ of this system in terms of angular frequency ' ω ' is given by $H(\omega)$
- (A) $\frac{1}{1+\omega}$ (B) $\frac{\sin \omega}{\omega}$
 (C) $\frac{1}{2+j\omega}$ (D) $\frac{3\omega}{2+\omega}$
7. The output of this system to the sinusoidal input $x(t) = 2 \cos(2t)$ for all time ' t ' is
- (A) 0
 (B) $2^{-0.25} \cos(2t - 0.125\pi)$
 (C) $2^{-0.5} \cos(2t - 0.125\pi)$
 (D) $2^{-0.5} \cos(2t - 0.25\pi)$
8. The impulse response of L.T.I system is $h(t) = e^{-a|t|}$, the step response is?
- (A) $\frac{1}{a}[e^{at} - e^{-at}u(t) - e^{at}u(t) + 2u(t)]$
 (B) $\frac{1}{a}[e^{-at} + e^{at}u(t) - e^{-at}u(t) - 2u(t)]$
 (C) $\frac{1}{a}[e^{at} + e^{-at}u(t) - e^{at}u(t) + 2u(t)]$
 (D) $\frac{1}{a}[e^{at} + e^{-at}u(t) - e^{at}u(t) - 2u(t)]$
9. Laplace transform of $(t^2 - 2t)u(t-1)$
- (A) $\frac{2e^{-s}(1-s)}{s^3}$ (B) $\frac{2e^s(1-s)}{s^3}$
 (C) $\frac{2e^{-s}(1+s)}{s^3}$ (D) $\frac{2e^s(1+s)}{s^3}$
10. The convolution of $x_1(t) = u(t+3)$ and $x_2(t) = \delta(t-5)$ is
- (A) $u(t)$ (B) $u(t-2)$ (C) $u(t+2)$ (D) $u(2)$
11. Laplace transform of $x(t) = \sin^2 tu(t)$ is
- (A) $\frac{4}{s(s^2+2)}$ (B) $\frac{2}{s(s^2+2)}$
 (C) $\frac{2}{(s^2+4)}$ (D) $\frac{2}{s(s^2+4)}$
12. The inverse Laplace transform of $\frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2+4}\right]$ is
- (A) $\cos^2 t$ (B) $\sin^2 t$
 (C) $\cos^2 2t$ (D) $\sin^2 2t$
- Direction for questions 13 and 14:**
13. The inverse Laplace transform of $\ln\left(\frac{s-a}{s-b}\right)$ is
- (A) $(e^{bt} - e^{at})u(t)$ (B) $\frac{1}{t}(e^{-bt} - e^{-at})u(t)$
 (C) $\frac{1}{t}(e^{bt} - e^{at})u(t)$ (D) $\frac{1}{t}(e^{-bt} + e^{-at})u(t)$
14. The inverse Laplace transform of $s \cdot \ln\left(\frac{s-a}{s-b}\right)$ is
- (A) $\frac{1}{t^2}[(bt-1)e^{bt} + (at-1)e^{at}] \cdot u(t)$
 (B) $\frac{1}{t^2}[(bt-1)e^{bt} - (at-1)e^{at}] \cdot u(t)$
 (C) $\frac{1}{t^2}[(bt+1)e^{bt} - (at+1)e^{at}] \cdot u(t)$
 (D) $\frac{1}{t^2}[(bt+1)e^{-bt} - (at-1)e^{-at}] \cdot u(t)$
15. Laplace transform of $e^{-3t}u(t) * tu(t)$
- (A) $\frac{-1}{s^2(s+3)}$ (B) $\frac{1}{s^2(s-3)}$
 (C) $\frac{1}{s^2(s+3)}$ (D) $\frac{-1}{s^2(s-3)}$
16. Poles and zeroes of $e^{2t}u(t) + e^{-3t}u(-t)$
- (A) Zero at $\frac{-5}{2}$, poles at 2, 3
 (B) Zero at $\frac{5}{2}$, poles at -2, -3
 (C) No common ROC
 (D) Zero at $\frac{-5}{2}$, poles at -2, -3
17. The initial and final values of $X(s) = \frac{2s}{s^2+2s+2}$ are
- (A) 2 and 1 (B) 2 and 0
 (C) 1 and 2 (D) 2 and 2
18. A system is formed by connecting two systems in cascade, the impulse response of these systems are $e^{-2t}u(t)$, and $2e^{-t}u(t)$. The impulse response of overall system is
- (A) $2(e^{-t} - e^{-2t})u(t)$ (B) $2(e^{-t} + e^{-2t})u(t)$
 (C) $(e^{-t} - e^{-2t})u(t)$ (D) $2(e^{-t} \cdot e^{-2t})u(t)$
19. $H(s)$ is the transfer function $H(s) = \frac{s}{s^2-s-2}$ ($\sigma < -1$). Then, function is
- (A) Causal and stable
 (B) Causal and unstable
 (C) Non-causal and stable
 (D) Non-causal and non-stable
20. A continuous time LTI system is described by $\frac{d^2}{dt^2}y(t) + 4\frac{d}{dt}y(t) + 3y(t) = 2\frac{d}{dt}x(t) + 4x(t)$. Assuming zero initial conditions, the response $y(t)$ of the above mentioned system for the input $x(t) = e^{-2t}u(t)$ is

- (A) $(e^{-t} - e^{-3t}) u(t)$
 (B) $(e^t + e^{3t}) u(t)$
 (C) $(e^{-t} + e^{-3t}) u(t)$
 (D) $(e^t - e^{3t}) u(t)$
21. A linear time-invariant, causal, continuous time system has a natural transfer function with sample poles at $s = -2$ and $s = -4$, and one sample zero at $s = -1$. A unit step $u(t)$ is applied at the input of the system.

At steady state, the output has constant value of 1. The impulse response of this system is

- (A) $-4e^{-2t} u(t)$ (B) $12e^{-4t} u(t)$
 (C) $(-4e^{-2t} + 12e^{-4t}) u(t)$ (D) None
22. Unit impulse response of a system is $f(t) = e^{-t}$, for $t \geq 0$. For this system, the steady state value of the output for unit step input is equal to.
- (A) 1 (B) -1 (C) 0 (D) ∞

Practice Problems 2

Direction for questions 1 to 17: Select the correct alternative from the given choices.

1. Match the function with ROC

P	$e^{-at} u(t)$	1.	$\sigma > 0$
Q	$-e^{-at} u(-t)$	2.	$\sigma > a$
R	$te^{at} u(t)$	3.	$\sigma < -a$
S	$t^n u(t)$	4.	$\sigma > -a$

- (A) P - 3, Q - 4, R - 2, S - 1
 (B) P - 4, Q - 3, R - 2, S - 1
 (C) P - 3, Q - 4, R - 1, S - 2
 (D) P - 4, Q - 3, R - 1, S - 2
2. The initial $x(0)$ and final value $x(\infty)$ of $X(s) = \frac{1}{s(s-1)}$ is
 (A) 1, 0 (B) 0, -1 (C) 2, 1 (D) 1, 2
3. If $x(t)$ and $X(s)$ are Laplace transform pairs, then Laplace transform of $e^{at} x(t)$ is
 (A) $e^{as} X(s)$ (B) $X(s)$
 (C) $X(s-a)$ (D) $X(s+a)$
4. If $x(t)$ and $X(s)$ are Laplace transform pairs, then Laplace transform of $tx(t)$ is
 (A) $\int_0^\infty X(s) ds$ (B) $\frac{-d}{ds} X(s)$
 (C) $\frac{\partial}{\partial s} X(s)$ (D) $X(s)e^{-s}$
5. Laplace transform of $\left[\frac{\sin \pi t}{t} \right] u(t)$ is
 (A) $\tan^{-1} \left(\frac{s}{\pi} \right)$ (B) $\cos^{-1} \left(\frac{s}{\pi} \right)$
 (C) $\cot^{-1} \left(\frac{s}{\pi} \right)$ (D) $\sin^{-1} \left(\frac{s}{\pi} \right)$
6. Laplace transform of a rectified half-sine wave with period 2π and amplitude '1' is
 (A) $\frac{(1+e^{-\pi s})}{s^2+1}$ (B) $\frac{(1-e^{-\pi s})}{s^2+1}$
 (C) $\frac{1}{(s^2+1)(1-e^{-\pi s})}$ (D) $\frac{1}{(s^2+1)(1+e^{-\pi s})}$

7. Laplace transforms of $x(t) = e^{at} u(t)$ and $y(t) = -e^{at} u(-t)$ have
 (A) same transforms but different ROCs.
 (B) different transforms but same ROCs.
 (C) same transforms and same ROCs.
 (D) different transforms and different ROCs.

8. Laplace transform and ROC of $e^{-5t} u(-t+3)$ is

- (A) $-\frac{e^{-3(s+5)}}{s+5}$, $\text{Re}(s) < 5$ (B) $\frac{e^{-3(s+5)}}{s+5}$, $\text{Re}(s) > 5$
 (C) $-\frac{e^{-3(s+5)}}{s+5}$, $\text{Re}(s) > 5$ (D) $\frac{e^{-3(s+5)}}{s+5}$, $\text{Re}(s) < 5$

9. Match the Laplace transform pairs

P.	$x(at)$	1.	$asX(s)$
Q.	$x(t-a)$	2.	$X(s-a)$
R.	$e^{at} \cdot x(t)$	3.	$e^{-sa} X(s)$
S.	$\frac{d}{dt}[ax(t)]$	4.	$\frac{1}{a} X\left(\frac{s}{a}\right)$, $a > 0$

- (A) P - 3, Q - 2, R - 1, S - 4
 (B) P - 4, Q - 2, R - 1, S - 3
 (C) P - 2, Q - 4, R - 3, S - 1
 (D) P - 4, Q - 3, R - 2, S - 1
10. Poles and zeroes of $x(t) = e^{-2t} u(t) + e^{-3t} u(t)$
 (A) Zero at $-\frac{5}{2}$, poles at 2, 3
 (B) Zero at $\frac{5}{2}$, poles at -2, -3
 (C) No common ROC
 (D) Zero at $-\frac{5}{2}$, poles at -2, -3
11. Inverse Laplace transform of $\frac{2s}{s+2}$
 (A) $2\delta(t) - 4e^{-2t}u(t)$
 (B) $2\delta(t) + 4e^{-2t}u(t)$
 (C) $2\delta(t) - 4e^{2t}u(t)$
 (D) $2\delta(t) + 4e^{2t}u(t)$
12. Given $H(s) = e^{-5s}$, what is the impulse response of the system?
 (A) $u(t-5)$ (B) $\delta(t-5)$
 (C) $e^{-5t} u(t)$ (D) $t \cdot e^{-5t} u(t)$

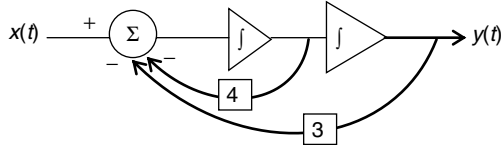
13. Convolution of $a^n u[n]$ and $b^n u[n]$ is

- (A) $\frac{b^{n+1} + a^{n+1}}{a+b} u[n]$ (B) $\frac{b^{n+1} - a^{n+1}}{b-a} u[n]$
 (C) $\frac{b^{n-1} - a^{n-1}}{b-a} u[n]$ (D) $\frac{b^{n-1} + a^{n-1}}{b-a} u[n]$

Direction for questions 14 and 15:

14. For a linear time-invariant system, whose block diagram is shown here with input $x(t)$ and output $y(t)$

The transfer function is

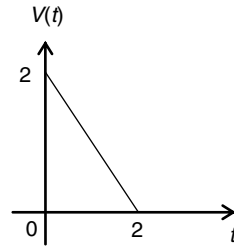


- (A) $\frac{1}{s^2 + 4s + 3}$ (B) $\frac{1}{s + 4}$
 (C) $\frac{1}{s - 3}$ (D) $\frac{1}{s^2 - 4s - 3}$

15. The step response of the system is

- (A) $\frac{-1}{2}e^{-t} + \frac{1}{6}e^{-3t}$ (B) $\frac{1}{3} + \frac{1}{2}e^{-t} - \frac{1}{6}e^{-3t}$
 (C) $\frac{1}{3} - \frac{1}{2}e^{-t} + \frac{1}{6}e^{-3t}$ (D) $\frac{1}{3} - \frac{1}{2}e^t + \frac{1}{6}e^{-3t}$

16. Express Laplace transform of



- (A) $\frac{-2}{s} + \frac{1}{s^2[1 - e^{-2s}]}$ (B) $\frac{2}{s} - \frac{1}{s^2}[1 + e^{-2s}]$
 (C) $\frac{-2}{s} - \frac{1}{s^2[1 - e^{-2s}]}$ (D) $\frac{2}{s} - \frac{1}{s^2}[1 - e^{-2s}]$

17. Given $L\{f(t)\} = \frac{s+2}{s^2+1}$, $L\{g(t)\} = \frac{s^2+1}{(s+3)(s+2)}$, $h(t) = \int_0^t f(\tau)g(t-\tau)d\tau$, then $L\{h(t)\}$ is

- (A) $\frac{1}{s+2}$ (B) $\frac{1}{s+1}$
 (C) $\frac{1}{s+3}$ (D) $\frac{s}{s+3}$

PREVIOUS YEARS' QUESTIONS

1. A causal system having the transfer function

$H(s) = \frac{1}{s+2}$ is excited with $10u(t)$. The time at which the output reaches 99% of its steady state value is [2004]

- (A) 2.7 s (B) 2.5 s (C) 2.4 s (D) 2.1 s

2. The unitstep response of a system starting from rest is given by

$$c(t) = 1 - e^{-2t} \text{ for } t \geq 0$$

The transfer function of the system is [2006]

- (A) $\frac{1}{1+2s}$ (B) $\frac{2}{2+s}$
 (C) $\frac{1}{2+s}$ (D) $\frac{2s}{1+2s}$

3. The unit impulse response of a system is $h(t) = e^{-t}$, $t \geq 0$.

For this system, the steady-state value of the output for unit step input is equal to [2006]

- (A) -1 (B) 0 (C) 1 (D) ∞

4. If the Laplace transform of a signal $y(t)$ is $Y(s) = \frac{1}{s(s-1)}$, then its final value is [2007]

- (A) -1 (B) 0
 (C) 1 (D) Unbounded

5. The 3 dB bandwidth of the low pass signal $e^{-t}u(t)$, where $u(t)$ is the unit step function, and it is given by [2007]

- (A) $\frac{1}{2\pi}$ Hz (B) $\frac{1}{2\pi}\sqrt{2}-1$ Hz
 (C) ∞ (D) 1 Hz

6. The frequency response of a linear, time-invariant system is given by $H(f) = \frac{5}{1+j10\pi f}$. The step response of the system is [2007]

- (A) $5(1 - e^{-5t})u(t)$ (B) $5\left(1 - e^{\frac{-t}{5}}\right)u(t)$
 (C) $\frac{1}{5}(1 - e^{-5t})u(t)$ (D) $\frac{1}{5}\left(1 - e^{\frac{-t}{5}}\right)u(t)$

7. A linear, time-invariant, causal continuous time system has a rational transfer function with simple poles at $s = -2$ and $s = -4$, and one simple zero at $s = -1$. A unit step $u(t)$ is applied at the input of the system. At steady state, the output has constant value of 1. The impulse response of this system is [2008]

- (A) $[\exp(-2t) + \exp(-4t)] u(t)$
 (B) $[-4\exp(-2t) + 12\exp(-4t) - \exp(-t)] u(t)$
 (C) $[-4\exp(-2t) + 12\exp(-4t)] u(t)$
 (D) $[-0.5\exp(-2t) + 1.5\exp(-4t)] u(t)$

Direction for questions 8 and 9:

The impulse response $h(t)$ of a linear time-invariant continuous time system is given by $h(t) = \exp(-2t) u(t)$, where $u(t)$ denotes the unit step function.

8. The frequency response $H(\omega)$ of this system in terms of angular frequency ω is given by $H(\omega) =$ [2008]

- (A) $\frac{1}{1 + j2\omega}$ (B) $\frac{\sin(\omega)}{\omega}$
 (C) $\frac{1}{2 + j\omega}$ (D) $\frac{j\omega}{2 + j\omega}$

9. The output of this system to the sinusoidal input $x(t) = 2\cos(2t)$ for all time t is [2008]

- (A) 0
 (B) $2^{-0.25} \cos(2t - 0.125\pi)$
 (C) $2^{-0.5} \cos(2t - 0.125\pi)$
 (D) $2^{-0.5} \cos(2t - 0.25\pi)$

10. An LTI system having transfer function $\frac{s^2 + 1}{s^2 + 2s + 1}$ and input $x(t) = \sin(t + 1)$ is in steady state. The output is sampled at a rate ω_s rad/s to obtain the final output $\{y(k)\}$. Which of the following is true? [2009]

- (A) $y(\cdot)$ is zero for all sampling frequencies ω_s
 (B) $y(\cdot)$ is non-zero for all sampling frequencies ω_s
 (C) $y(\cdot)$ is non-zero for $\omega_s > 2$, but zero for $\omega_s > 2$
 (D) $y(\cdot)$ is zero for $\omega_s > 2$ but non-zero for $\omega_s > 2$

11. A system with transfer function $\frac{Y(s)}{X(s)} = \frac{s}{s + p}$ has an output $y(t) = \cos\left(2t - \frac{\pi}{3}\right)$ for the input signal $x(t) = p \cos\left(2t - \frac{\pi}{2}\right)$. Then, the system parameter 'p' is [2010]

- (A) $\sqrt{3}$ (B) $\frac{2}{\sqrt{3}}$ (C) 1 (D) $\frac{\sqrt{3}}{2}$

12. Given $f(t) = L^{-1}\left[\frac{3s + 1}{s^3 + 4s^2 + (K - 3)s}\right]$. If $\lim_{t \rightarrow \infty} f(t) = 1$, then the value of K is [2010]
 (A) 1 (B) 2 (C) 3 (D) 4

13. A continuous time LTI system is described by $\frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 3y(t) = 2 \frac{dx(t)}{dt} + 4x(t)$. Assuming zero initial conditions, the response $y(t)$ of the abovementioned system for the input $x(t) = e^{-2t}u(t)$ is given by [2010]

- (A) $(e^t - e^{3t})u(t)$ (B) $(e^{-t} - e^{-3t})u(t)$
 (C) $(e^{-t} + e^{-3t})u(t)$ (D) $(e^t + e^{3t})u(t)$

14. The trigonometric Fourier series of an even function does not have the [2011]

- (A) dc term (B) cosine terms
 (C) sine terms (D) odd harmonic terms

15. An input $x(t) = \exp(-2t)u(t) + \delta(t - 6)$ is applied to an LTI system with impulse response $h(t) = u(t)$. The output is [2011]

- (A) $[1 - \exp(-2t)] u(t) + u(t + 6)$
 (B) $[1 - \exp(-2t)] u(t) + u(t - 6)$
 (C) $0.5[1 - \exp(-2t)] u(t) + u(t + 6)$
 (D) $0.5[1 - \exp(-2t)] u(t) + u(t - 6)$

16. If $F(s) = L[f(t)] = \frac{2(s + 1)}{s^2 + 4s + 7}$, then the initial and final values of $f(t)$ are, respectively, [2011]

- (A) 0, 2 (B) 2, 0
 (C) 0, 2/7 (D) 2/7, 0

17. The unilateral Laplace transform of $f(t)$ is $\frac{1}{s^2 + s + 1}$. The unilateral Laplace transform of $tf(t)$ is [2012]

- (A) $-\frac{s}{(s^2 + s + 1)^2}$ (B) $-\frac{2s + 1}{(s^2 + s + 1)^2}$
 (C) $\frac{s}{(s^2 + s + 1)^2}$ (D) $\frac{2s + 1}{(s^2 + s + 1)^2}$

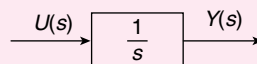
18. The impulse response of a system is $h(t) = tu(t)$. For an input $u(t - 1)$, the output is [2013]

- (A) $\frac{t^2}{2} u(t)$ (B) $\frac{t(t - 1)}{2} u(t - 1)$
 (C) $\frac{(t - 1)^2}{2} u(t - 1)$ (D) $\frac{t^2 - 1}{2} u(t - 1)$

19. Which one of the following statements is NOT TRUE for a continuous time causal and stable LTI system? [2013]

- (A) All the poles of the system must be on the left-hand side of the $j\omega$ -axis.
 (B) Zeroes of the system can be anywhere in the s -plane.
 (C) All the poles must be within $|s| = 1$
 (D) All the roots of the characteristic equation must be located on the left-hand side of the $j\omega$ -axis.

20. Assuming zero initial condition, the response $y(t)$ of the following system to a unit step input $u(t)$ is [2013]



- (A) $u(t)$ (B) $tu(t)$
 (C) $\frac{t^2}{2} u(t)$ (D) $e^{-t}u(t)$

21. A system is described by the differential equation

$$\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y(t) = x(t).$$

Let $x(t)$ be a rectangular pulse given by

$$x(t) = \begin{cases} 1, & 0 < t < 2 \\ 0, & \text{otherwise} \end{cases}$$

Assuming that $y(0) = 0$ and $\frac{dy}{dt} = 0$ at $t = 0$, the transform of $y(t)$ is [2013]

- (A) $\frac{e^{-2s}}{s(s+2)(s+3)}$ (B) $\frac{1-e^{-2s}}{s(s+2)(s+3)}$
(C) $\frac{e^{-2s}}{(s+2)(s+3)}$ (D) $\frac{1-e^{-2s}}{(s+2)(s+3)}$

22. A continuous, linear time-invariant filter has an impulse response $h(t)$ described by

$$h(t) = \begin{cases} 3 & \text{for } 0 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

When a constant input of value 5 is applied to this filter, the steady state output is _____. [2014]

23. A system described by the following differential equation, where $u(t)$ is the input to the system and $y(t)$ is the output of the system.

$$\dot{y}(t) + 5y(t) = u(t)$$

When $y(0) = 1$ and $u(t)$ is a unit step function, $y(t)$ is

- (A) $0.2 + 0.8e^{-5t}$ (B) $0.2 - 0.2e^{-5t}$
(C) $0.8 + 0.2e^{-5t}$ (D) $0.8 - 0.8e^{-5t}$

24. Let $h(t)$ denote the impulse response of a causal system with transfer function $\frac{1}{s+1}$. Consider the following three statements.

S1: The system is stable

S2: $\frac{h(t+1)}{h(t)}$ is independent of t for $t > 0$

S3: A non-causal system with the same transfer function is stable. [2014]

For the abovementioned system,

- (A) only S1 and S2 are true
(B) only S2 and S3 are true
(C) only S1 and S3 are true
(D) S1, S2, and S3 are true

25. A stable linear time-invariant (LTI) system has a transfer function $H(s) = \frac{1}{s^2 + s - 6}$. To make this system causal, it needs to be cascaded with another LTI system having a transfer function $H_1(s)$. A correct choice for $H_1(s)$ among the following options is [2014]

- (A) $s+3$ (B) $s-2$ (C) $s-6$ (D) $s+1$

26. A causal LTI system has zero initial conditions and impulse response $h(t)$. Its input $x(t)$ and output $y(t)$ are related through the linear constant-coefficient differential equation [2014]

$$\frac{d^2 y(t)}{dt^2} + \alpha \frac{dy(t)}{dt} + \alpha^2 y(t) = x(t)$$

Let another signal $g(t)$ be defined as

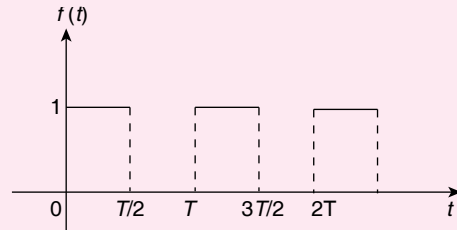
$$g(t) = \alpha^2 \int_0^t h(\tau) d\tau + \frac{dh(t)}{dt} + \alpha h(t)$$

If $G(s)$ is the Laplace transform of $g(t)$, then the number of poles of $G(s)$ is _____.

27. Let the signal $f(t) = 0$ outside the interval $[T_1, T_2]$, where T_1 and T_2 are finite. Furthermore, $|f(t)| < \infty$. The region of convergence (RoC) of the signal's bilateral Laplace transform $F(s)$ is [2015]

- (A) a parallel strip containing the $j\Omega$ axis
(B) a parallel strip not containing the $j\Omega$ axis
(C) the entire s -plane
(D) a half plane containing the $j\Omega$ axis

28. The Laplace transform of the causal periodic square wave of period T shown in the figure below is: [2016]



- (A) $F(s) = \frac{1}{1 + e^{-\frac{ST}{2}}}$ (B) $F(s) = \frac{1}{s \left(1 + e^{-\frac{ST}{2}} \right)}$
(C) $F(s) = \frac{1}{s(1 - e^{-ST})}$ (D) $F(s) = \frac{1}{1 - e^{-ST}}$

29. A first order low pass filter of time constant T is excited with different input signals (with zero initial conditions up to $t = 0$). Match the excitation signals X, Y, Z with the corresponding time responses for $t \geq 0$: [2016]

- X: Impulse P: $1 - e^{-\frac{t}{T}}$
Y: Unit step Q: $t - T \left(1 - e^{-\frac{t}{T}} \right)$
Z: Ramp R: $e^{-\frac{t}{T}}$

- (A) $X \rightarrow R, Y \rightarrow Q, Z \rightarrow P$
(B) $X \rightarrow Q, Y \rightarrow P, Z \rightarrow R$
(C) $X \rightarrow R, Y \rightarrow P, Z \rightarrow Q$
(D) $X \rightarrow P, Y \rightarrow R, Z \rightarrow Q$

ANSWER KEYS

EXERCISES

Practice Problems 1

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. B | 4. C | 5. D | 6. C | 7. D | 8. A | 9. A | 10. B |
| 11. D | 12. A | 13. C | 14. B | 15. C | 16. C | 17. B | 18. A | 19. D | 20. A |
| 21. C | 22. A | | | | | | | | |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|------|------|-------|
| 1. B | 2. B | 3. C | 4. B | 5. C | 6. C | 7. A | 8. A | 9. D | 10. D |
| 11. A | 12. B | 13. B | 14. A | 15. C | 16. D | 17. C | | | |

Previous Years' Questions

- | | | | | | | | | | |
|-------|--------------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. C | 4. D | 5. A | 6. B | 7. C | 8. C | 9. D | 10. A |
| 11. B | 12. D | 13. B | 14. C | 15. D | 16. B | 17. D | 18. C | 19. C | 20. B |
| 21. B | 22. 44 to 46 | | 23. D | 24. A | 25. B | 26. 1 | 27. C | 28. B | 29. C |